VISHNU INSTITUTE OF TECHNOLOGY :: BHIMAVARAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FULL STACK DEVELOPMENT- I

(SKILL ENHANCEMENT COURSE)

PROJECT REPORT

Project Title: Vitual Queue

DEPARTMENT : Computer Science and Engineering **SECTION : D**

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PROJECT REPORT

Project Overview

The QueueLess is a web-based virtual queue management system designed to eliminate physical waiting lines. The application allows users to join a queue remotely using a QR code or a unique queue code. Once joined, users can track their live position in the queue along with an estimated waiting time (ETA).

The system is mainly useful in environments such as college cafeterias, clinics, and service centers where long queues are common. Admins can efficiently manage the queue by serving or skipping users in real time. The application improves time management, reduces crowding, and provides a smooth and organized queue experience.

Additionally, the system ensures fairness by maintaining proper queue order and prevents overcrowding through capacity control. It also enhances user convenience by allowing them to utilize their waiting time productively instead of standing in physical lines.

Key Features

1. Feature One: (Queue Creation & Management)

- Admin can create a queue with name, capacity, and service time
- Generates a unique code or QR for users to join
- Allows real-time queue control
- Supports dynamic queue updates without page refresh
- Ensures controlled entry using capacity limits.

2. Feature Two: (Live Queue Tracking)

- Users can join queue using code or QR
- Displays real-time position in queue
- Shows estimated waiting time (ETA)
- Automatically updates position when queue changes
- Provides a clear and simple interface for tracking.

3. Feature Three: (Smart Notifications & Control)

- Notifies user when their turn is near
- “You’re Next” alert for position 1
- Admin can Serve Next or Skip users
- Helps users prepare in advance without constant checking
- Reduces missed turns and improves efficiency

Technologies Used

Technology / Tool	Purpose / Description
React (Vite)	Frontend framework for building dynamic and responsive UI
Firebase Auth	Provides secure anonymous authentication for users
Firestore	Real-time database to store and sync queue data instantly
JavaScript	Handles application logic and dynamic interactions
HTML & CSS	Used for structuring and styling the application interface
Firebase Hosting	Deploys and hosts the application efficiently

Project Structure

The project is organized into three main sections: Home, User Queue, and Admin panel. The Home screen allows users to either create or join a queue. The User Queue screen displays the user's current position and estimated waiting time with live updates. The Admin panel allows managing the queue by serving or skipping users.

The frontend is built using modular React components, ensuring maintainability and reusability. Firebase services handle authentication and real-time database operations efficiently. The architecture is designed to be simple, scalable, and easy to extend in the future.

User Flow

The application follows a simple and intuitive user flow:

- User opens the application
- Chooses to join or create a queue
- If joining, enters queue code or scans QR
- User gets assigned a position in queue
- User tracks position and ETA live
- Admin manages queue using Serve Next / Skip
- User receives notification when turn is near
- Queue updates automatically without manual refresh.

Database Design(If Used)

Collection / Table	Data Stored
Queues	Queue name, max capacity, avg service time, status
Users (subcollection)	User ID, position, join time
Queue Info	Current size, pause status

Improvement Ideas

1. Add real-time push notifications using Firebase Cloud Messaging
2. Improve UI animations and transitions
3. Add user authentication with login options
4. Enable multi-queue support
5. Add reminder notifications
6. Add estimated waiting accuracy improvements
7. Improve error handling and validations
8. Add admin analytics dashboard
9. Introduce queue history tracking
10. Add dark mode support for better UI experience

Future Scope

The project can be extended with advanced features:

- Mobile app version using Flutter
- AI-based wait time prediction
- Integration with payment systems
- Multi-location queue management
- Advanced notification system
- Role-based admin access
- Integration with smart devices or kiosks

References

Source Code (GitHub): https://github.com/Harishthokka/vitual_queue.git

Project Deployed URL: https://virtual-queue.lovable.app